

Project 1 – Abnormal EV Charging Sessions

Task 1: Data Understanding, Audit and Preparation

MSc in Informatics Engineering
Data Mining (MINDD)



Students:

Amir Masnavi | 1221579@isep.ipp.pt

Ines Oliveira | 1231205@isep.ipp.pt

Manuela Leite | 1200720@isep.ipp.pt

Professor: Alberto Pinheira
Academic year: 2026/2027

Abstract

In this first task we studied 441,077 electric-vehicle charging sessions. The goal of the project is to predict, when a session starts, whether it will end abnormally. The file has no target column, so we built it from the end cause: the 13 causes that are faults count as abnormal, which gives 18.58% abnormal sessions. We checked this definition against the dataset's own history columns, and they match in about 100% of the rows.

Several columns are only known after the session ends (energy, costs, end and payment times, end cause), so we removed them from the predictors, together with the order creation time. We found no duplicates or invalid records. The missing values have a clear reason and the extreme values are real, so we did not remove any row. Volume, abnormal share and type of fault change over the two years, so we split the data by time and not at random.

We keep 29 features in 7 groups. The recent failure history of the charging post was the strongest signal we found.

Contents

1	Introduction	1
1.1	Target definition	1
2	Data Audit and Exploration	1
2.1	Availability at prediction time	1
2.2	Data quality	2
2.3	What relates to abnormal termination	2
2.4	Changes over time	3
3	Preparation	4
3.1	Chronological split	4
3.2	Features	5
3.3	Learned operations and class imbalance	5
4	Synthesis	6
	Bibliography	7

List of Figures

2.1	Univariate ROC-AUC on the training period, coloured by availability at the start	2
2.2	Abnormal share by decile of six history variables (dashed line: training average)	3
2.3	Monthly volume, abnormal share, new users and active posts	4

Acronyms

AUC	Area Under the Curve
EV	Electric Vehicle
IQR	Interquartile Range
NaN	Not a Number (missing value)
PCA	Principal Component Analysis
PR-AUC	Area under the Precision–Recall curve
ROC-AUC	Area under the Receiver Operating Characteristic curve
SMOTE	Synthetic Minority Over-sampling Technique
VIF	Variance Inflation Factor

1 Introduction

The project asks whether the information available when an Electric Vehicle (EV) charging session starts can predict that the session will end abnormally. We work with an educational version of the China EV Charging Dataset [1]: 441,077 sessions and 47 columns, from 31 Dec 2019 to 31 Dec 2021, at 92 charging posts in 13 stations and 3 districts. All the numbers in this report come from our notebook `Task1.ipynb` (seed 42).

We followed three rules during the whole task. First, a variable can only be a predictor if it is known when the session starts. Second, before fixing a problem in the data, we tried to understand it. Third, any value learned from the data (for example, the mean used for scaling) is learned from the training period only.

1.1 Target definition

The brief describes the target `is_Abnormal`, but the file does not include it. It only has `end_cause`, with 15 values. We decided that a session is *normal* when it ends with “Charging ends normally” or “User stops charging”, because stopping is the user’s own choice. The other 13 causes are equipment, communication or vehicle faults, so we label them *abnormal*.

To check this choice, we used the history columns of the dataset. For each session, they count how many earlier sessions of the same post or user were abnormal. We recounted them with our label and with an alternative where only “Charging ends normally” is normal. Our definition reproduces the supplied counts almost exactly, and the alternative does not (Table 1.1). This means the dataset was built with the same definition as ours.

Table 1.1: Candidate target definitions and agreement with the supplied history columns

Definition of “normal”	Post, 30 days	User, 30 days	Abnormal share
A: ends normally + user stops	100.00%	99.98%	18.58%
B: ends normally only	0.26%	35.85%	53.93%

With this definition there is about 1 abnormal session for every 4.4 normal ones. A model that always predicts “normal” would already be 81% accurate, so accuracy alone is not a good metric here. We also noticed two details. The supplied rates are smoothed as $(\text{abnormal} + 1) / (\text{sessions} + 5)$, so a post or user with no history always gets the rate 0.2. In addition, 9,058 normal sessions (2.5%) delivered no energy, mostly users who stopped after about one minute. We kept them as normal, because removing them would require the energy, which is only known after the session.

2 Data Audit and Exploration

2.1 Availability at prediction time

Seven columns describe what happened during the session: energy, electricity cost, service charge, amount, actual payment, end time and payment time. `end_cause` is the outcome itself. Figure 2.1 shows why these columns are a problem. On its own, the energy, each money column and the duration separate normal and abnormal sessions with an Area

under the Receiver Operating Characteristic curve (ROC-AUC) of 0.82–0.85, better than any variable known at the start (best 0.77). The payment delay is weaker (0.61), but it is also only known after the session. The reason is that abnormal sessions are short and deliver little energy, so these columns describe the result instead of predicting it, so all of them were excluded.

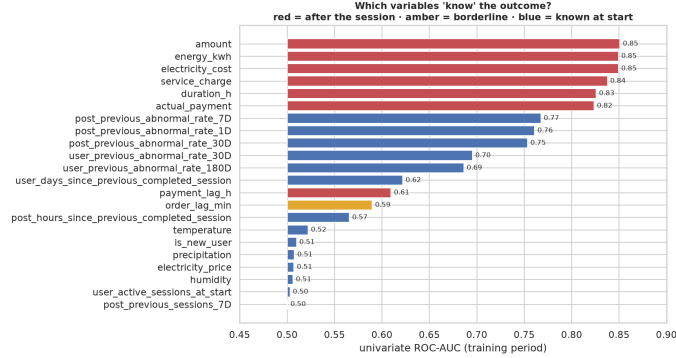


Figure 2.1: Univariate ROC-AUC on the training period, coloured by availability at the start

The order creation time was a harder case. It is stamped after the start in 56% of the sessions (57.2% in the training period, sometimes 16 h later), so we cannot be sure it exists when we make the prediction. Its link to the outcome is also weak (AUC 0.59), so we excluded it too. Weather has only one value per district and day. Following the brief, we treat it as a forecast that is known at the start.

2.2 Data quality

Table 2.1 lists the checks we made. In general, the records are in good condition. The missing values appear only when there was no earlier session or failure, so they are not really unknown. For the outliers, we first used the Interquartile Range (IQR) rule and then looked at the extreme cases one by one. They turned out to be real situations, so we did not remove any row.

2.3 What relates to abnormal termination

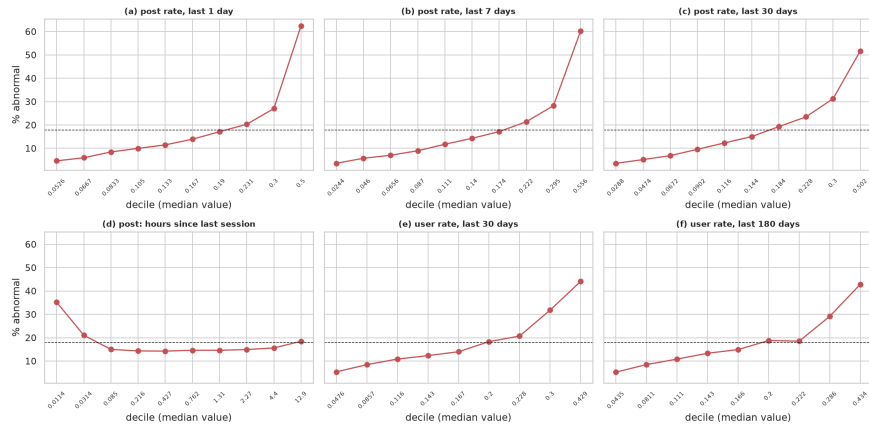
We measured these relationships on the training period only, so that the validation and test periods do not influence our choices. Location matters, but mostly at the level of the post. The abnormal share is 14.6–24.1% per district and about 8–38% per station, while the posts range from 4.1% to 60.9%. The 10 worst posts produce 24.1% of the abnormal sessions with only 10.2% of all sessions. Calendar and weather effects are small. Across the hours of the day the abnormal share moves between 14.7% and 20.5%, across the days of the week between 17.0% and 18.9%, and the weather effects are weak and not linear.

The history columns gave the clearest signal (Figure 2.2). When we split the 7-day post rate into deciles, the abnormal share goes from about 4% in the lowest decile to about 60% in the highest. In other words, a post that failed recently tends to fail again. User history also helps, but less: its AUC is around 0.69–0.70, against 0.75–0.77 for post history.

The users are very unequal. In the training period, 45% of the 34,294 accounts have only one session, but the top 1% of accounts have 34% of the sessions. New users (12% of

Table 2.1: Data-quality checks, findings and decisions

Check	Finding	Decision
Formatting	Trailing tab in the order and payment times and in 18,370 location values	Stripped on load
Duplicates	No identical rows; 17 pairs where one account starts on two posts in the same second (fleet accounts)	Kept
Consistency	No negative values and no session ending before it starts; durations stop at 23.99 h; 54,621 sessions with zero energy	Kept (these are after-session fields)
Missing values	Only in 4 “time since previous” columns; every NaN matches its “no history” flag	Not deleted, not filled with the mean (Section 3.3)
Hidden codes	No −999 or “unknown” codes; values of 999 and above are real counts of fleet accounts	No action
Outliers (IQR)	Long right tails in counts and time-since columns (skewness 3–9)	Kept; $\log_{1p}$ for linear models
Outliers (checked)	Posts idle for more than 14 days (39% abnormal on return), 21 accounts with more than 500 sessions, two storm days; the sessions flagged by Isolation Forest are 60% abnormal	Real and informative, so kept

**Figure 2.2:** Abnormal share by decile of six history variables (dashed line: training average)

the sessions) fail a little more often than returning users (20.2% against 17.5%).

2.4 Changes over time

Figure 2.3 shows that the data are not stable over the two years. Monthly volume drops to 1,391 sessions in February 2020, probably because of COVID-19, and reaches 25–28k

sessions per month in autumn 2021. The abnormal share falls from 26.3% to 13.7% and later rises again to 22.9%. The type of fault also changes: connection faults add 7.1 percentage points to the abnormal share in Q1-2020, but only 1–2 points from Q3-2020 onwards. At the start of the file the history columns are still incomplete, which is why 31% of the sessions in January 2020 come from “new” users, against about 9–15% later. Finally, posts 17–20 stop on 28 Sep 2021. Some things are stable, though. The abnormal share of each post is similar in 2020 and 2021 (Spearman 0.81), which explains why post history works. Because of all the changes above, a random split would mix different periods and give results that are too optimistic, so we split the data by time.

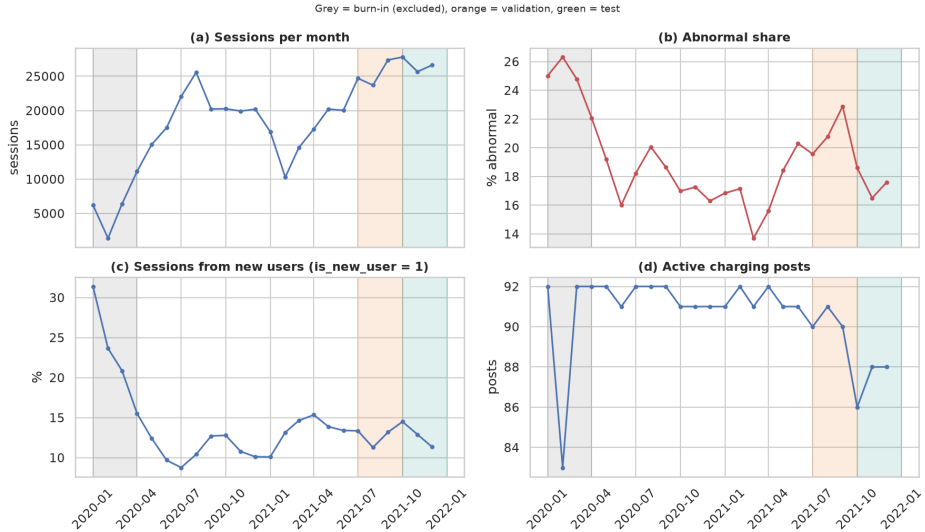


Figure 2.3: Monthly volume, abnormal share, new users and active posts

3 Preparation

3.1 Chronological split

We did not use the sessions before 1 April 2020 (the *burn-in*), because their history columns are incomplete and their fault mix no longer exists. The remaining data were split by start date into training, validation and test periods (Table 3.1). The label of a session is only known when it ends, so we also removed (*purged*) the 28 training sessions that end after the training cut-off.

Table 3.1: Chronological split

Period	Dates	Sessions	Abnormal
Burn-in (not used)	31 Dec 2019 to 31 Mar 2020	14,017	25.0%
Train	1 Apr 2020 to 30 Jun 2021	271,182	17.8%
Purged	Train starts that end after 1 Jul 2021	28	–
Validation	Jul to Sep 2021	75,769	21.1%
Test (used once)	Oct to Dec 2021	80,081	17.6%

The split shows two things that matter for Task 2. First, 39.9% of the validation sessions and 61.3% of the test sessions come from accounts that never appear in training,

so `user_id` cannot be a feature. Second, every post in validation and test was already seen in training, so we cannot test a brand-new post. For tuning, we will use expanding-window folds inside the training period instead of a random K-fold.

3.2 Features

We keep 29 features in 7 groups: calendar (hour and day of week), tariff period, location (`post_id`, station and district), weather (temperature, humidity, precipitation and a new `rain_flag`), post history (8), user history (10) and concurrency (1). Table 3.2 lists the start-time columns we dropped.

Table 3.2: Start-time columns that were dropped

Column	Reason
<code>electricity_price</code>	Fixed by the tariff period
Six abnormal-count columns	Exact functions of the session counts and rates
Two post “no history” flags	Always 0 after the burn-in
<code>user_no_previous_completed_session</code>	Same as <code>is_new_user</code> in 99.9% of rows
<code>user_id</code>	Key field with 56k values; most test accounts are new

Some features are strongly related to each other. Only the user-history windows have a Variance Inflation Factor (VIF) above 10 (up to 65.4), because the 180-day and 730-day windows contain almost the same sessions. Tree models and regularised linear models can deal with this, so they get all the features. For plain linear or distance-based models we built a smaller set. We went through the features from the best to the worst univariate AUC and kept a feature only if its $|\rho|$ with the features already kept was below 0.90. This leaves 19 of the 23 numeric features and brings the maximum VIF down to 8.6.

Principal Component Analysis (PCA) was also tested. It needs 11 of the 23 components to keep 90% of the variance, and the first component hardly separates the classes (AUC 0.51). Since it compresses little and would mix post and user information, we did not use it. Discretisation was only used to explore the data (deciles and weather bands), not as a model input.

3.3 Learned operations and class imbalance

Different models need different preparation, so we prepared two versions of the data. The tree version keeps the raw values, the NaN and the category codes. The linear version applies `log1p` to 8 skewed columns, fills the structural NaN with the training maximum (we read “never happened” as “a very long time ago”) and keeps the flags. It then standardises the numeric columns, encodes hour and weekday with sine and cosine, and one-hot encodes `post_id` and the tariff period. The result has 126 columns and no NaN.

All operations that learn something are fitted on the training period only, inside a scikit-learn pipeline (Table 3.3). We checked what would happen if we fitted them on all periods instead, and the learned values change. For example, the fill value for “days since the user’s last failure” would move from 543 to 702 days, which would leak information from the validation and test periods.

Table 3.3: Learned operations, all fitted on the training period

Operation	What is learned	Applies to
One-hot encoding / category codes	List of categories	Linear / tree version
Structural-NaN fill	Training maximum (flags kept)	Linear version
Standardisation	Mean and standard deviation	Linear version
Log-transform rule	Which columns are skewed	Linear version (8 columns)
Reduced feature set	Spearman ρ and AUC (training sample)	Plain linear / distance models
Class-imbalance treatment	Class weights or resampling	Training folds only (Task 2)
Outlier capping, PCA, rare-category grouping	Nothing	Not used / not needed

The classes (about 1 : 4.4) are not balanced in the prepared data. If Task 2 uses class weights or resampling, they will be applied only inside the training folds, so validation and test keep their real distribution. We will be careful with Synthetic Minority Over-sampling Technique (SMOTE), because interpolating between sessions can create values that cannot exist, such as half a flag or a mix of two posts.

4 Synthesis

The most important thing we learned is that the outcome is easy to “predict” with the wrong columns. Energy, cost and duration separate the classes better than anything known at the start, but only because they describe the result. Once these columns are removed, the best signal is the recent failure history of each post. Location and user history add some information, while calendar and weather add little. Our second main finding is that the data change over time, in volume, in abnormal share and in the type of fault. The records were in good condition, so no row was removed or imputed, and we keep 29 features in 7 groups (19 numeric features for plain linear models).

Some questions remain open. We do not know whether the daily weather values were a forecast or were measured during the day; in the second case they would contain information from after the start. Durations stop at 24 h, and we could not explain the timing of the order and payment fields. Our explanations for the changes over time, such as COVID-19, are hypotheses. We also cannot test a new post, and user history is a behaviour profile, which raises a privacy question.

For Task 2, we will evaluate the models on the time split, use the test period only once and report precision, recall and Area under the Precision–Recall curve (PR-AUC) against the base rate of each period. We plan to start with tree ensembles and a regularised logistic regression, and to run the feature-group ablation with and without `post_id` and user history.

Bibliography

- [1] Zhang, Y., Xu, T., Chen, T., et al. (2025). A high-resolution electric vehicle charging transaction dataset with multidimensional features in China. *Scientific Data*, 12, 643. <https://doi.org/10.1038/s41597-025-04982-1>